

Pete Cowles

SOFTWARE / STRUCTURAL ENGINEER

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github.com/KilgoreTrout9 

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SKILLS

Languages

JavaScript | HTML |
Python

Front End

React | Leaflet | Redux |
React Router | CSS |
Material-UI

Back End

NodeJS | Express |
Kubernetes | Docker |
NginX | MySQL |
PostgreSQL | MongoDB |
Mongoose

Technologies

Jira | Asana | Github |
NPM | AWS (EC2) |
Postman

CERTIFICATIONS

FAA DAS AR

Bombardier Aerospace
2003

EDUCATION

Hack Reactor

Software Engineering
Immersive 2021

Univ. of Illinois - Champaign

Bachelor of Science
Aero/Astro Engineering
1991

SOFTWARE EXPERIENCE

DRAPER LABS – Full Stack Software Engineer

2022 - 2023

- Created new API resources for in-situ teams to coordinate movements and deliveries.
- Employed **OpenLayers** within **Typescript** modules to display lat/long and MGRS coordinates and allow users to select new icons for CSV imported map events.

BRILLIO – Full Stack Software Engineer

2021-2022

- Designed new visualization method using **Leaflet** to display heatmaps, coverage maps and cell towers relating to telecommunication network events.
- Optimized legacy **JavaScript** and **NodeJS** code base to utilize automated testing in **Jest** and improved automated processing with **Docker** and **Kubernetes**.
- Coordinated user stories and network testing for off-shore groups with **Postman**.

GALVANIZE – Software Engineer in Residence

2021

- Mentoring and coaching full stack software engineers in **JavaScript**, **React**, **NodeJS**, and **DBMSs**.
- Facilitated the coordinated group creation of front-end and back-end application using **AWS**, **Nginx**, **Docker** and **Jest**.

SOFTWARE APPLICATIONS

Smove

github.com/BlueOcean-Smove/Smove

- Developed a web based local moving service to client standards via user stories and live demos.
- Maintained unique customer login and friend groups with **Google Auth 2.0**.

ReviewIt

github.com/KilgoreTrout9/MVP-Movie_Rater

- Designed a full stack, web based, editable movie review collection utility with **JavaScript**, **React**, **Redux**, **Material-UI** and **MongoDB** with **Mongoose**.
- Persisted data utilizing **MongoDB** and **Mongoose** and deployed to the Atlas cloud service.

Catwalk E-Commerce Platform

github.com/Agile-Ankylosaurus/Project-Catwalk

- Designed mock-ups, codebase, demos and user stories for an e-commerce website specializing in the user ratings and reviews using **JavaScript**, **React**, **HTML5** and **CSS3**.
- Achieved Lighthouse ratings of: Performance – 97, Accessibility – 92, SEO – 80.
- Horizontally scaled the **AWS EC2** deployment using **Docker** and **Nginx** to process server loading: 2500 req/s, latency < 400 ms, 0% errors.

ENGINEERING EXPERIENCE

Altitude Aerospace – Lead Contract Engineer

2023-Present

- Managed a group of 6 engineers including scheduling deliverables and project planning.
- Analyzed and reviewed analysis packages for FAA submittal.

LMI Aerospace Inc – Aircraft Sustaining Program Manager

2014-2018

- Managed a group of 7 engineers including scheduling deliverables and project planning.
- Provided on time FAA approval for after-market airplane STC installations and off-site repairs.

Bombardier Aerospace – Global Express Stress Group Lead

1999-2004

- Coordinated and approved stress group reports to support on-time airplane delivery schedule
- Analyzed new installations and structural reports as basis for FAA approval

Senior Structural Contract Engineer

1991-2018

- Developed a redesign for the 767 winglet to correct Damage Tolerance design issues
- Analyzed repairs for a B-747 P2C conversion that was approved by FAA
- Redesigned the nose landing gear for increased loading on the new Syberjet SJ30-2